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FROM LEARNING TO BUILDING

10 DAYS OF INNOVATION



TEAM 04



Anushka
Raghav



Rucha
Musale



Prachi
Tewari



Dhananjay
Chimnani



OUR MISSION

To transform classroom learning into a production-ready enterprise application by mastering modern technologies, collaborating as a team, and building ReconX—a Trade Reconciliation Platform.

DAY 1



LEARNING

DAY 3



BUILDING

DAY 5



TESTING & SECURITY

DAY 7



FRONTEND DEVELOPMENT

DAY 10



DEPLOYMENT & PRESENTING



TOGETHER, WE LEARNED.



TOGETHER, WE BUILT.



TOGETHER, WE DELIVERED.



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THE PROBLEM WE SET OUT TO SOLVE

Challenges in Traditional Trade Reconciliation



1. Manual Reconciliation

- Time-consuming
- Error-prone



2. Scattered Data

- Multiple disconnected systems
- Difficult tracking



3. Security Challenges

- Weak authentication
- Limited auditability



4. Limited Visibility

- Delayed insights
- Slow decisions



Trade Execution



Multiple Systems



Manual Verification



Errors & Delays



Operational Risk



ReconX



Automated
Reconciliation



Secure
Authentication



Role-Based
Access



Real-Time
Monitoring



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BUILDING RECONX

Transforming Enterprise Learning into a Complete Solution



IMPLEMENTATION & KEY FEATURES

Built in iterations using Agile methodology with a focus on reliability, scalability and observability.

KEY FEATURES



AUTOMATED TRADE RECONCILIATION

Automatically matches trades, detects breaks and publishes reconciliation results.



SECURE AUTHENTICATION

JWT based authentication and role based access control for secure operations.



AUDIT & TRACEABILITY

Complete audit trail for all trades and reconciliation events.



REAL-TIME NOTIFICATIONS

Alerts and notifications for mismatches and critical events.



DASHBOARDS & REPORTS

Interactive dashboards for monitoring trades, breaks and system health.



SCALABLE & RESILIENT

Event-driven architecture ensures high throughput and fault tolerance.

IMPLEMENTATION FLOW



PRESENTATION LAYER



React + Vite

- Login & Authentication
- Trade Management
- Reconciliation Dashboard
- Reports & Monitoring



API LAYER



Spring Boot

- REST APIs
- Business Logic
- Reconciliation Engine
- Audit Service



INTEGRATION LAYER



Apache Kafka

- Event Streaming
- Trade Events
- Reconciliation Results
- Audit Events



DATA LAYER



PostgreSQL

- Trades & Reference Data
- Reconciliation Results
- Audit Logs
- User & Role Management



OBSERVABILITY LAYER



Prometheus



Grafana



Spring Actuator

Metrics Collection • Dashboards • Alerts • Health Checks



DEVELOPMENT APPROACH

Agile (Iterative & Incremental)



SPRINTS COMPLETED

6+ Sprints



TEST CASES EXECUTED

250+



DEFECTS RESOLVED

95%+



UAT APPROVAL

100%



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THE JOURNEY. THE LEARNINGS. THE IMPACT.

Key Takeaways & What's Next



KEY LEARNINGS



Collaboration & Communication
Aligned as a team, delivered as one.



Security & Reliability
Built secure, robust and audit-ready by design.



Continuous Learning
Explored new technologies and solved real-world challenges.



IMPACT DELIVERED



Reduced reconciliation time and manual effort



Improved accuracy, transparency and auditability



Stronger security with role-based access



WHAT'S NEXT



AI-powered matching and anomaly detection



Cloud-native deployment and scalability



Advanced analytics and dashboards

“

From Challenge to Impact.
We built more than a platform — we built capability.

”



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THANK YOU

Team 04 | Built with Purpose. Learned for the Future.



Grateful for the opportunity.

Excited for what's ahead.